

ECE 315

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Lecture 30 :

- Motion control
- Stepper Motors
- HW controllers, H-bridge



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Last Lecture

- Analogue/Digital conversion
- ADCs, DACs

This Lecture

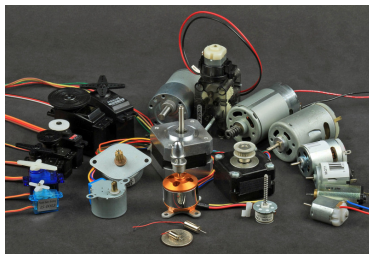
- Motion control
- Stepper motors
- HW controllers, H-bridge

Learning Objectives

- Explain the role of electromechanical actuators
- Describe the operating principle of a stepper motor
 - Explain full-step and half-step operation
 - Interpret H-bridge operation
- Explain the motivation for sophisticated stepper motor driver ICs, such as the TMC2209
- Describe and illustrate two ways to use a microcontroller to drive the TMC2209
 - What microcontroller peripherals might be useful (e.g., UART and...?)

Electro-mechanical Motion and Control

- Up to now we have focused on interfaces to things that sense or indicate change
- Can also interface to devices that change the environment in some way, act on something, such as motors, servos, and actuators



Actuators, Relays, Servos and Stepper Motors

- Microcomputers are often required to control the movement of mechanical devices or to set other physical parameters.
- An **actuator** is a device that is used by a microcomputer to control an external physical quantity, such as position.
- A **relay** is an electrical switch that is closed and opened by driving or not driving current through a magnetic coil.
 - Normally open (n.o.) relay: close the switch by energizing
 - Normally closed (n.c.) relay: open the switch by energizing
- A **servo motor** is an actuator that is used to cause mechanical rotation within some fixed range of angles (<360 deg.).
- A **stepper motor** is an actuator whose rotational position (360 deg.) and speed can be controlled by digital signals.

Stepper Motors

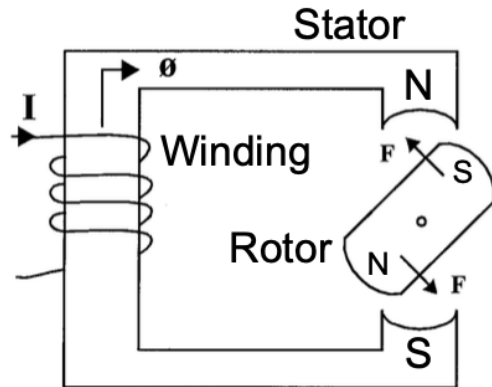
- An electromechanical actuator that converts a sequence of electrical control pulses into discrete, incremental angular motion
 - Each input pulse causes the motor shaft to rotate by a fixed step angle, allowing precise control of position and speed without requiring feedback

Key characteristics

- **Discrete motion:** Rotation occurs in fixed steps (e.g., 1.8° per step $\rightarrow$ 200 steps/revolution).
- **Open-loop control:** Position is determined by counting steps; no encoder is required in basic operation.
- **High holding torque:** Can maintain position when energized.
- **Repeatable positioning:** Accuracy depends on step angle and load conditions.

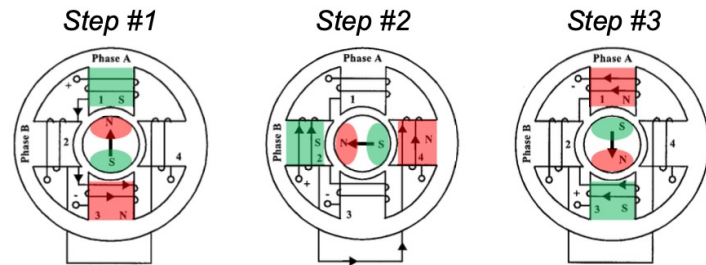
Stepper Motor Operation – Simplified

- A rotating rotor contains a permanent magnet with fixed north (N) and south (S) poles
- A stationary stator is energized with current to produce an electro-magnet
 - The location of the N and S poles of the electromagnet is determined by the current direction through the 1 or more winding(s)
- The rotor experiences a torque that acts to bring the rotor into alignment with the stator (the N and S poles attract one another)
 - The torque disappears when the stator is fully aligned with the rotor

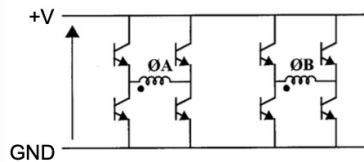


Stepper Motor Operation cont'd

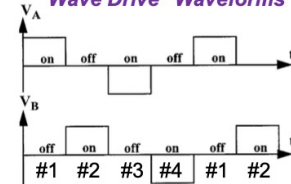
- Each transistor in the H-bridge can be controlled
- They are switched to shunt current one or the other way through the coil (A or B)



Dual Bipolar Full H-Bridge Circuit



"Wave Drive" Waveforms

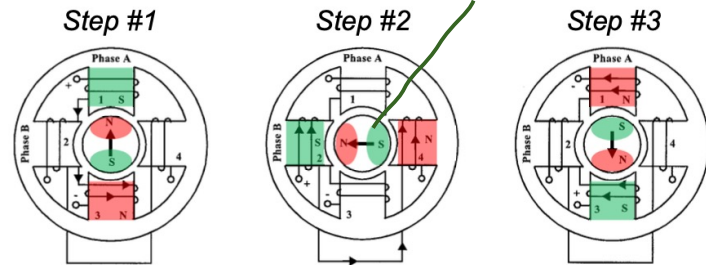


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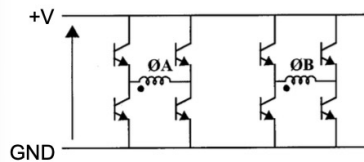
Stepper Motor Operation cont'd

Round rotor is a permanent magnet

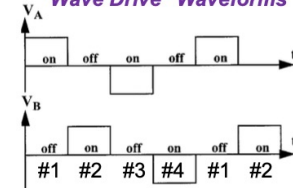
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Dual Bipolar Full H-Bridge Circuit



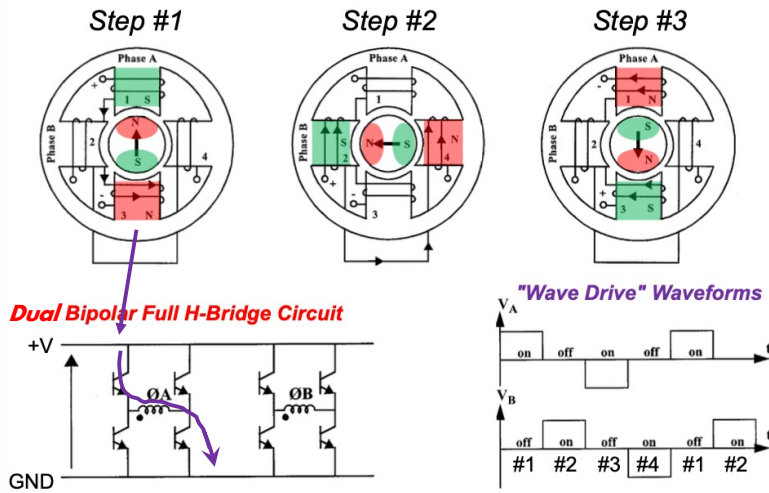
"Wave Drive" Waveforms



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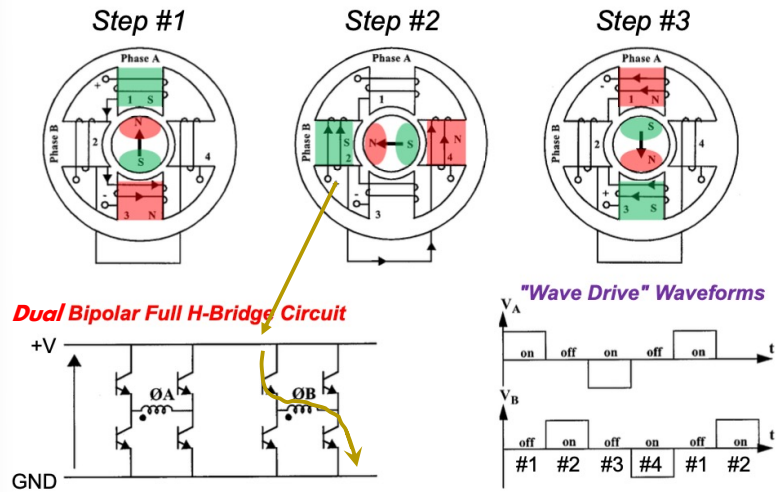
Stepper Motor Operation – Full Step

- Step #1 is the starting position
- Current flows into the “top” of coil A



Stepper Motor Operation – Full Step

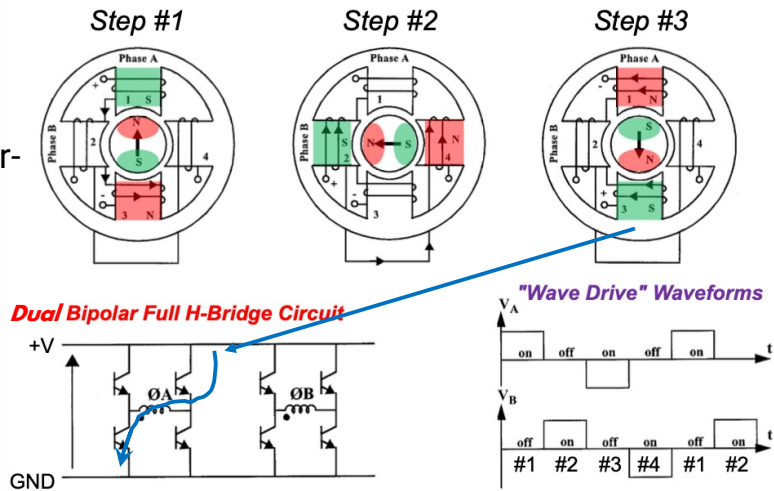
- In step #2, current flows into the top of coil B
- The rotor rotates counter-clockwise 90°



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Stepper Motor Operation – Full Step

- In step #3, current flows into the bottom of coil A
- The rotor rotates counter-clockwise another 90°
- In reality, no stepper motor has steps this coarse.
 - Standard step is 1.8° or less
 - $360/1.8 = 200$ steps



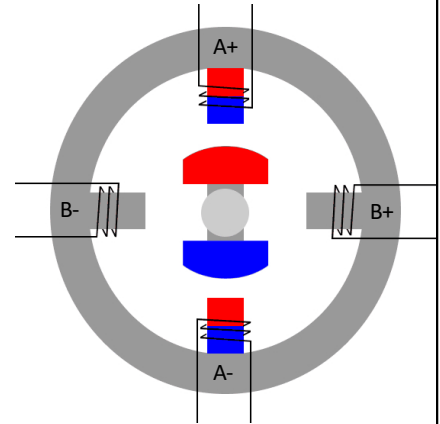
Stepper Motor Operation – Half Step

- For full steps, we energize the coils independently
- What if after energizing coil A in one step, we keep it energized and apply current to coil B?

Stepper Motor Operation – Half Step

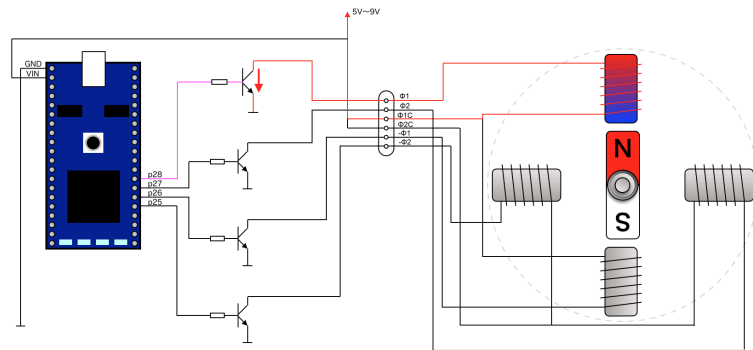
- For full steps, we energize the coils independently
- What if after energizing coil A in one step, we keep it energized and apply current to coil B?
- Now both coils are energized the same and will both pull on the same pole of the rotor
- The rotor will be equally attracted by both coils and settle in between them

$A+ \rightarrow A+B+ \rightarrow B+ \rightarrow B+A- \rightarrow A- \rightarrow A-B- \rightarrow B- \rightarrow B-A+ \rightarrow A+ \rightarrow \dots$

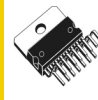


Stepper Motor Operation – Half Step

- A simplified version of a half-step driver circuit
- The microprocessor can't sink the high current needed by the motor, so uses transistors to handle the power



L298 Dual (Full) H-Bridge Driver IC



Multiwatt15 V

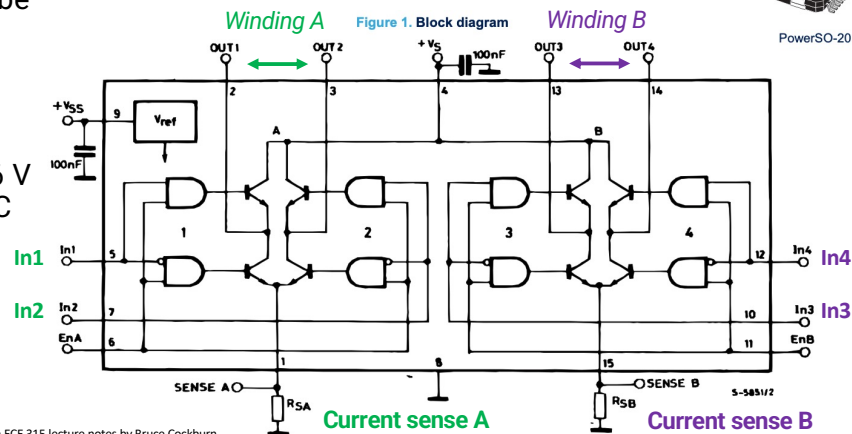


Multiwatt15 H



PowerSO-20

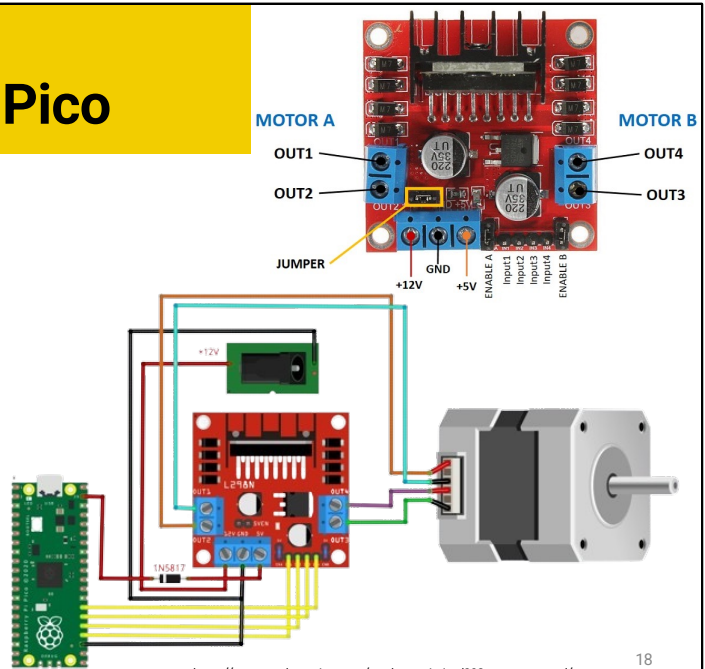
- TTL-compatible control inputs. Can be driven by many microcontrollers
- The H-bridge drives inductive loads (motors) at up to 46 V and a total of 4 A DC



L298N Controller + Pico

Forward half-step sequence

Step	In1	In2	In3	In4	Motor A	Motor B
1	H	L	L	L	+	
2	H	L	H	L	+	+
3	L	L	H	L		+
4	L	H	H	L	-	+
5	L	H	L	L	-	
6	L	H	L	H	-	-
7	L	L	L	H		-
8	H	L	L	H	+	-

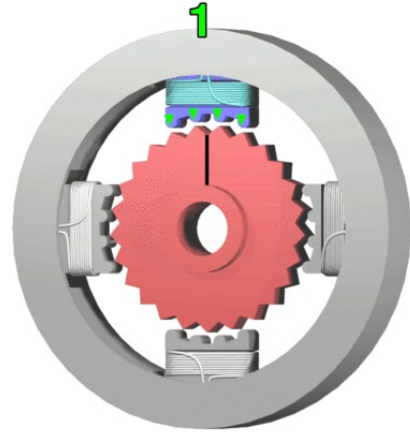


<https://www.teachmemicro.com/raspberry-pi-pico-l298n-motor-control/>

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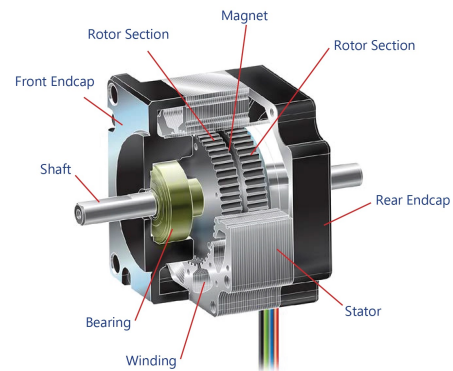
Smaller steps

- As mentioned before, a typical step is 1.8°
- This is accomplished by constructing the rotor to have more fixed magnets that are uniformly distributed
- The animation shows a full-step
- Half-step can be done just the same as shown on a previous slide



Hybrid Stepper Motor

- A hybrid stepper has a multi-pole stator and a two-part rotor
- The rotor comprises two physically identical multi-tooth permanent magnets, offset by a half tooth
- They are magnetized along the shaft axis, one north the other south
- The stator coils are wound on alternating poles with corresponding teeth



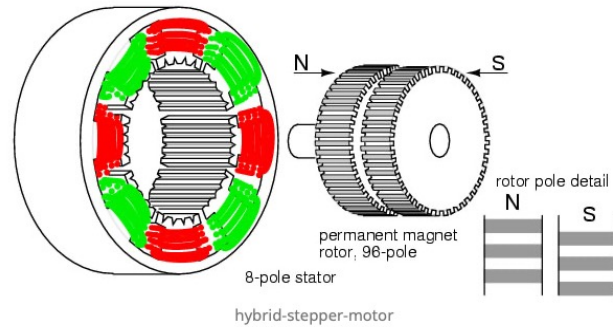
<https://www.hsmagnets.com/blog/hybrid-stepper-motor/>

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<https://www.linengineering.com/technology/hybrid-stepper-motors?srsId=AfmBOoqYKnHkkSjZm8ZXfU6-kfnNUp0HU9UZpEeQKbqvmzJN67kRcSUy>

Hybrid Stepper Motor cont'd

- The 48 teeth on one rotor disk are offset by half a tooth with respect to the other disk
- Due to the offset, the rotor effectively has 96 interleaved poles of opposite polarity
- This allows for a step of $1/96^{\text{th}}$ of a revolution by reversing the phase of a winding
- With two stator phases, there are 2×96 steps per revolution
- This means that the step angle is $360^\circ / 192 = 1.875^\circ$



Sophisticated Motor Drivers – TMC2209

- Stepper motors are used in many, many applications
 - 3D printers
 - Computer Numerically Controlled (CNC) machines
 - Robotics
 - Printers and scanners
 - Camera control systems
- This justifies the development of sophisticated driver/controllers that have built-in control algorithms and circuits
- Let's look at one very popular driver, the TMC2209

Sophisticated Motor Drivers – TMC2209

- From the manufacturer's (TRINAMIC) data sheet
The TMC2209 is an ultra-silent motor driver IC for two phase stepper motors (...) [The] StealthChop2 chopper ensures noiseless operation, maximum efficiency, and best motor torque
- **StealthChop2™** No-noise, high-precision chopper algorithm for inaudible motion and inaudible standstill of the motor. Supports very fast motor acceleration and deceleration.
- **SpreadCycle™** High-precision cycle-by-cycle current control for highest dynamic movements.
- **MicroPlyer™** Microstep interpolator for obtaining full 256 microstep smoothness with lower resolution step inputs starting from fullstep
- **StallGuard4™** Sensorless homing safes end switches & warns in case of motor overload
- **CoolStep™** Uses StallGuard measurement to adapt the motor current for best efficiency and lowest heat-up of motor and driver



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<https://www.analog.com/en/products/tmc2209.html>

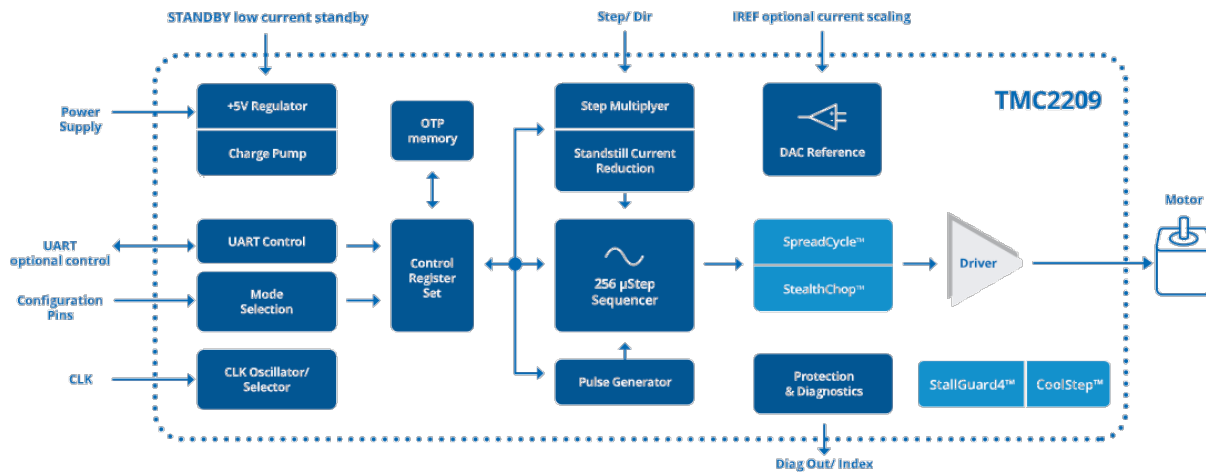
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Sophisticated Motor Drivers – TMC2209

- The chip provides two interfaces
 - STEP/DIR; A CPU (μ C) generates step & direction signals synchronized to additional motors and other components within the system. The TMC2209 operates the motor as commanded by the configuration pins & STEP/DIR signals
 - UART; uni- or bi-directional
The μ C sends commands to the controller via the UART interface; supports
 - Detailed diagnostics and thermal management
 - Passive braking and freewheeling for flexible, lowest power stop modes
 - More options for microstep resolution setting (fullstep to 256 microstep)
 - Software controlled motor current setting and more chopper options
 - Use StallGuard for sensorless homing and CoolStep for adaptive motor current and cool motor



Sophisticated Motor Drivers – TMC2209



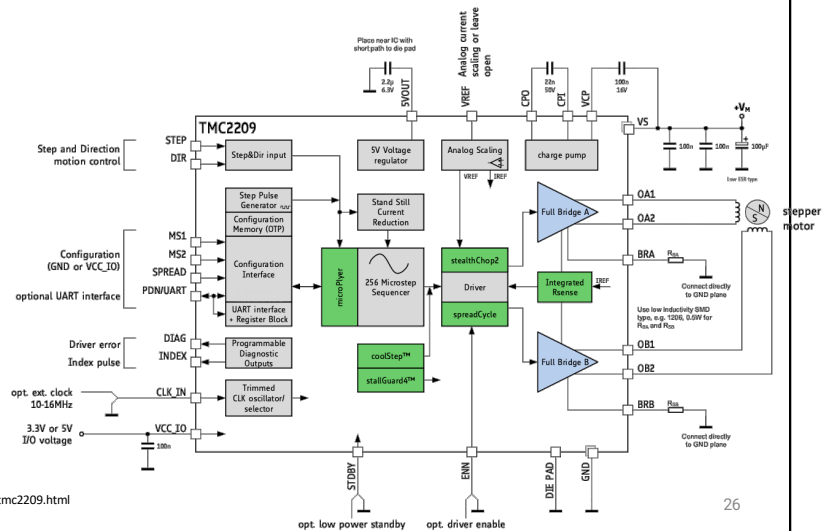
From the block diagram we see this is a pretty sophisticated chip.

Of note is

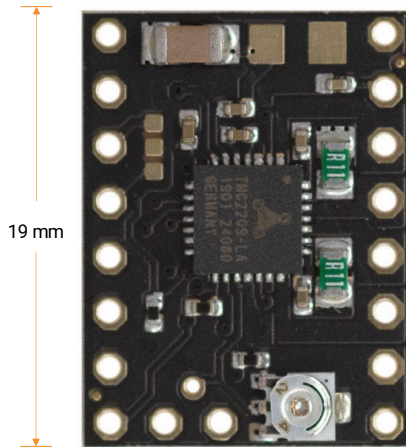
- the UART controller
- OTP, one-time programmable memory
- The register bank
- The digital to analogue converter (DAC) that is used as part of the control loop
- Custom circuits to provide advanced capabilities

Sophisticated Motor Drivers – TMC2209

- A little more detail including the pins
- Notice the minimal number of external components needed
 - 7 capacitors
 - 2 resistors
- From this a dual-inline pin module can be made

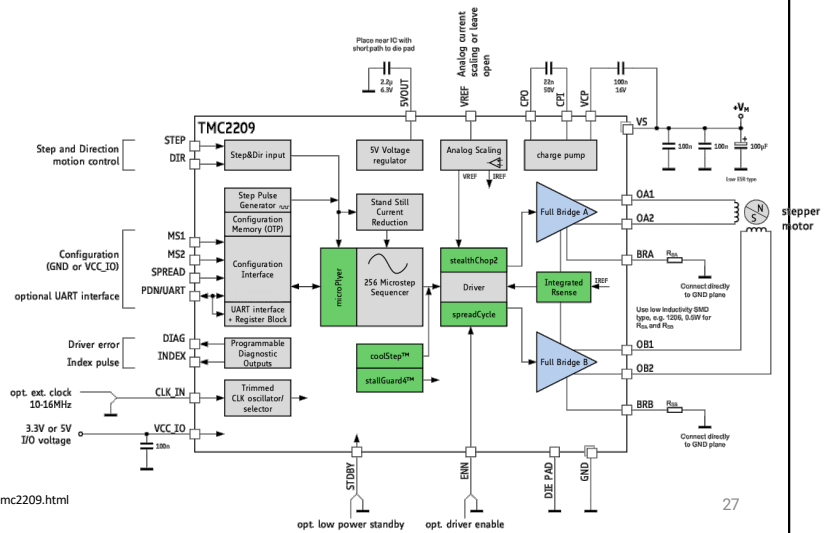


Sophisticated Motor Drivers – TMC2209

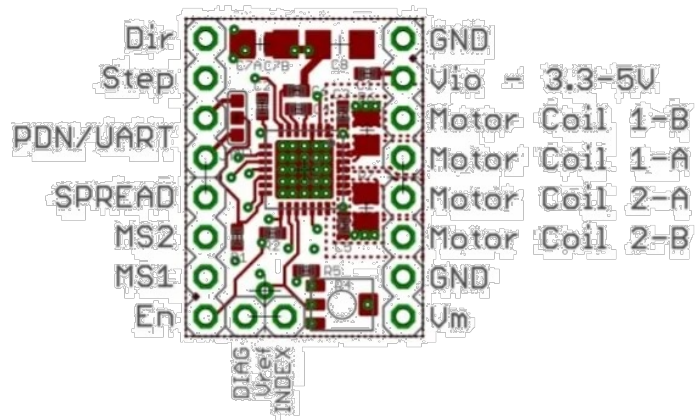
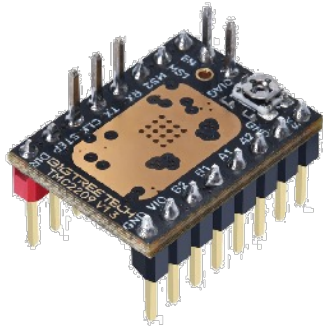


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Sophisticated Motor Drivers – TMC2209

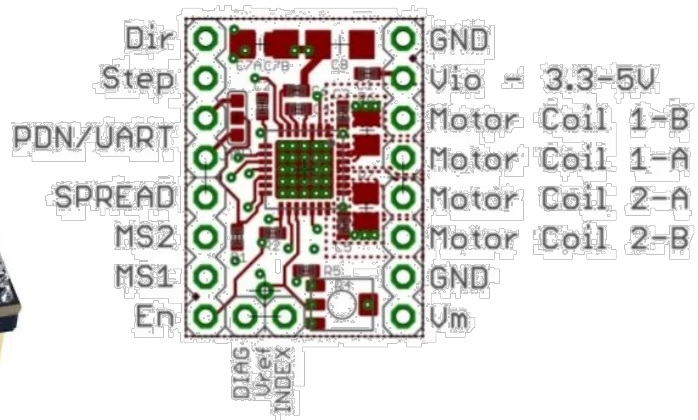
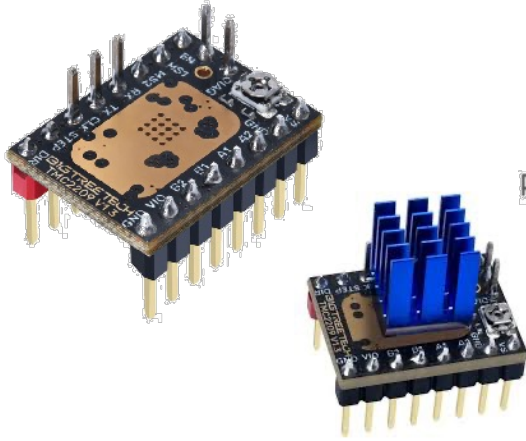


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<https://www.analog.com/en/products/tmc2209.html>

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Sophisticated Motor Drivers – TMC2209



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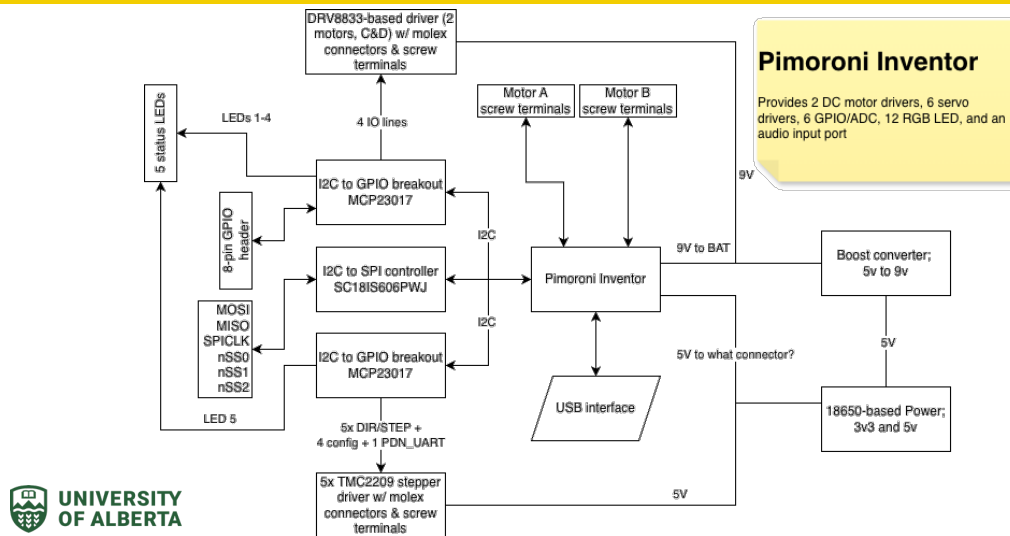
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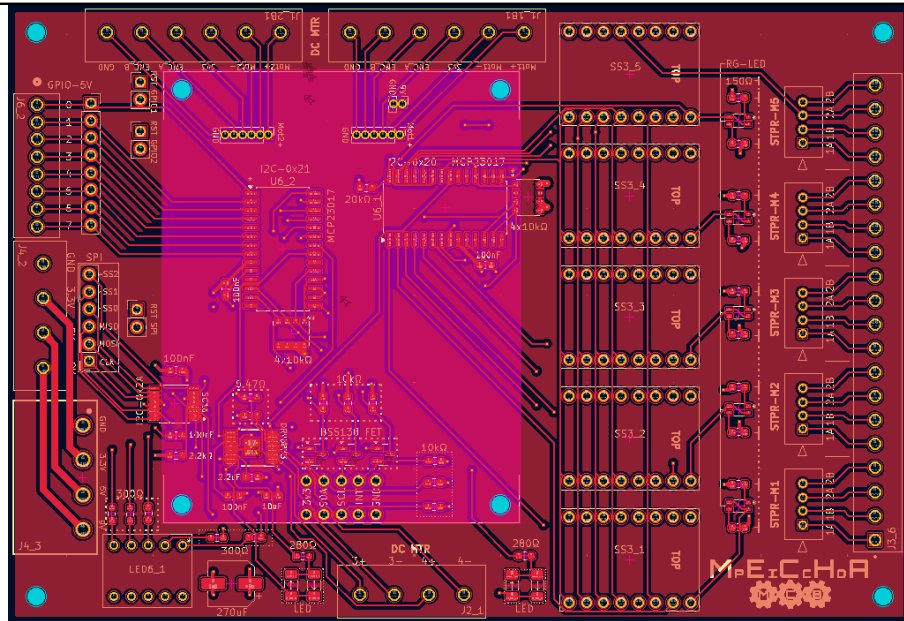
Mechatronics Motor Controller Board

- To support the Mechatronics and Robotics labs, a custom motor controller/sensors board has been developed
 - Raspberry Pi Pico-based (RP2040)
- Can control
 - 6 servos
 - 5 steppers
 - 4 DC motors
- Has
 - 12 RGB LEDs
 - 5 status LEDs
 - SPI with 3 chip selects
 - I2C
 - 8 + 6 GPIOs, 3 ADC capable



MCTR Motor Controller/Sensors Board





Summary

- Embedded systems can initiate and control physical motion using actuators such as relays, servos, DC motors, and stepper motors
- Stepper motors provide precise, repeatable positioning by converting digital pulse sequences into discrete angular steps (open-loop control)
- Full-step and half-step drive modes determine step resolution and smoothness through controlled coil energization sequences
- Special construction of stepper stator and rotors support small angle steps
 - Typically, a single step is 1.8° . Micro-stepping supports even finer steps
- H-bridge circuits enable bidirectional current flow in motor windings and are essential for microcontroller-based motor control
- Modern driver ICs (e.g., TMC2209) integrate current regulation, microstepping, diagnostics, & communication interfaces (STEP/DIR, UART) to simplify & enhance motion control system design

Next Lecture

- Motion control
- Encoders
- Servo and DC Motors
- Other actuators

Questions

